

Speech and Audio Processing

05 - speech recognition

Shinnosuke Takamichi / 高道 慎之介



Takamichi Laboratory
慶應義塾大学 高道研究室

Class schedule (tentative)

01	Oct. 03	Introduction	Learn from basic to state-of-the-art about speech processing.
02	Oct. 10	Speech production / 音声生成	
03	Oct. 17	Speech synthesis / 音声合成	
04	Oct. 24	Speech synthesis 2 / 音声合成2	
05	Oct. 31	Speech recognition / 音声認識	
06	Nov. 7	Self-supervised learning / 自己教師あり学習	
07	Nov. 14	Sound recognition/synthesis / 環境音認識合成	
08	Nov. 28	Foundation model / 基盤モデル	
09	Dec. 05	Advanced topics / 発展的課題	
10	Dec. 12	Speech system building / 音システム構築	Learn how to build your speech system.
11	Dec. 19	Speech system building / 音システム構築	
12	Dec. 26	Speech system building / 音システム構築	
13	Jan. 09	Speech system building / 音システム構築	
14	Jan. 23	Final presentation / 最終発表	

About this class

- Language
 - Slides in English (w/ partial translation to Japanese, e.g., named entity)
 - Talk in Japanese (w/ simultaneous translation to English.)
 - Once translation finished for current slide, I will move onto next slide.
 - Either En or Ja is fine for your question.
- Grades
 - 50 / 100 ... final presentation
 - 50 / 100 ... final report
- Room
 - 14-202

Final presentation & reports / 最終発表&レポート

- Perform a unique project that either
 - improves on state-of-the-art in speech-related {signal processing, machine learning} tasks,
 - applies {machine learning, signal processing} to a unique task,
 - investigates human behaviors in human–speech-AI interaction.
 - prototypes the future of human and speech AI.
- Presentation & report
 - Slides in English.
 - Talk in Japanese or English. (choose one you prefer.)
 - I will drive {Ja, En}-to-{En, Ja} simultaneous translation during presentation.
 - Reports in Japanese or English. (choose one you prefer.)

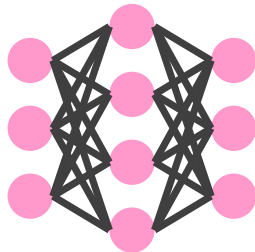
Previous class: acoustic models / 音響モデル

Modeling relationship between linguistic-acoustic features.

$$P(\overline{\mathbf{y}}|\overline{\mathbf{x}}) \approx \underbrace{P(\mathbf{y}|\mathbf{o})P(\mathbf{o}|\mathbf{l})}_{\text{Prob. of speech feats. given by linguistic feats. (acoustic model / 音響モデル)}} P(\mathbf{l}|\mathbf{x})$$

Speech features Textual features

Speech Text



Today's goal: automatic recognition (speech to text)

Mapping speech into text.

**Expanding
speech
recognition**

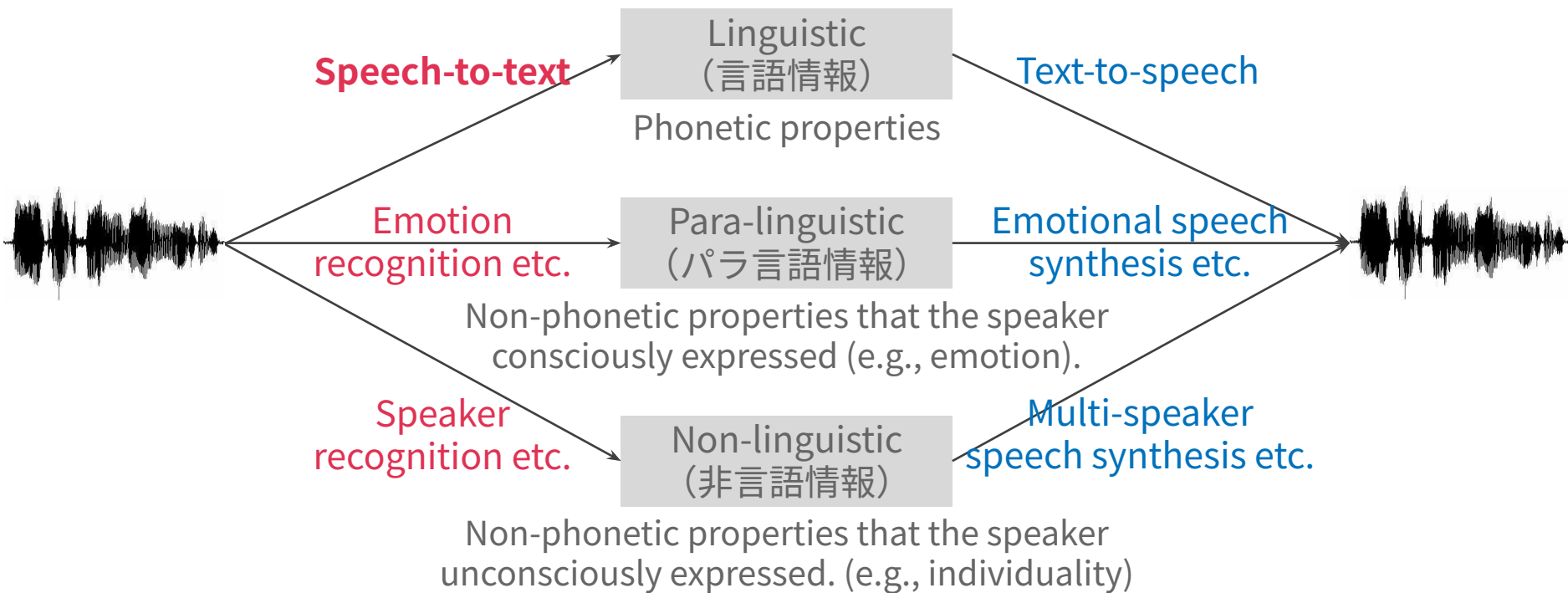
Google Research



Formulation of speech recognition / 音声認識の定式化

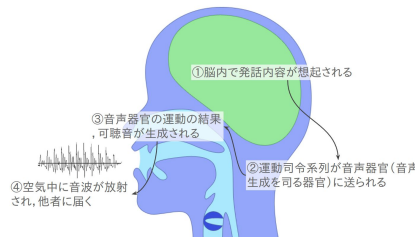
Speech information

Physical → Information → Physical



Formulation / 定式化

- A method to artificially recognize spoken text.
 - Emulating humans' speech recognition by computers



- **Described as generative models (生成モデル)**

- Why generative models? -> To be robust to randomness.
- Randomness in
 - Speaker: speech production, emotion, disease, accented language, etc.
 - Channel: background environments, telecommunication, reverberation, etc.



Mitigating into subproblems / 部分問題への緩和

$P(\mathbf{w}|\mathbf{o})$ Original formulation corresponding to “end-to-end” modeling.

$$= \sum_l P(\mathbf{w}, l | \mathbf{o})$$

Introducing intermediate linguistic feats l .

$$= \sum_l \frac{P(\mathbf{o} | \mathbf{w}, l) P(l | \mathbf{w}) P(\mathbf{w})}{P(\mathbf{o})}$$

Bayes' theorem

$$\propto \sum_l \frac{P(\mathbf{o} | \mathbf{w}, l) P(l | \mathbf{w}) P(\mathbf{w})}{P(\mathbf{o})}$$

Denominator (分母) is independent on $\mathbf{w}$

$$= \sum_l P(\mathbf{o} | l) P(l | \mathbf{w}) P(\mathbf{w})$$

Speech is independent on words.

Mitigating into subproblems / 部分問題への緩和

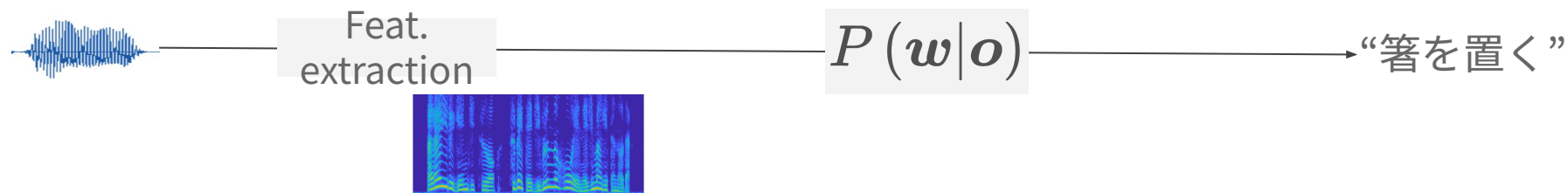
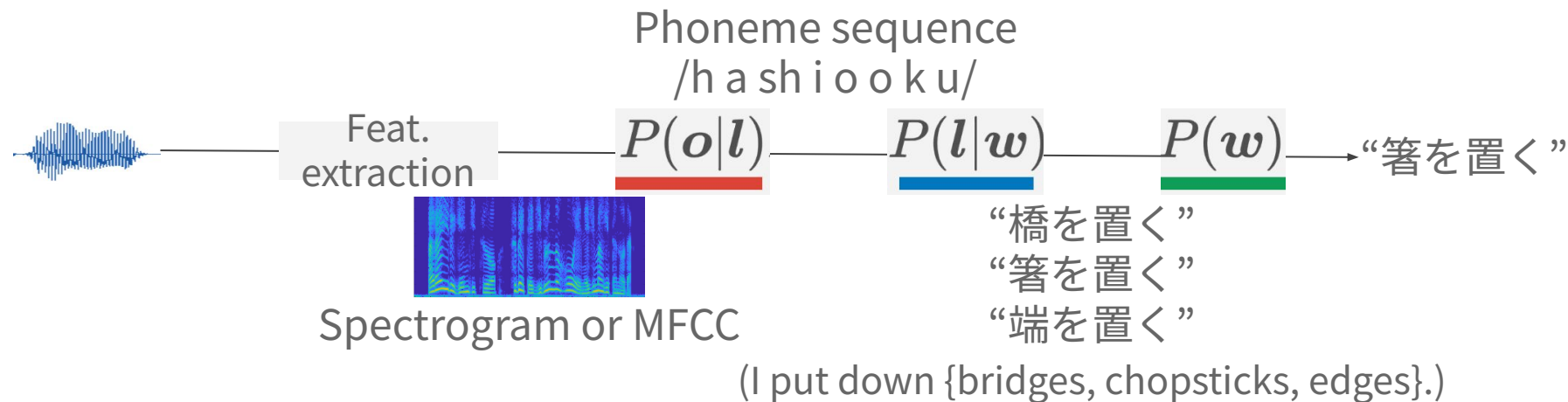
$$P(o|l)P(l|w)P(w)$$

Prob. of speech feats.
given by linguistic feats.
(acoustic model
/ 音響モデル)

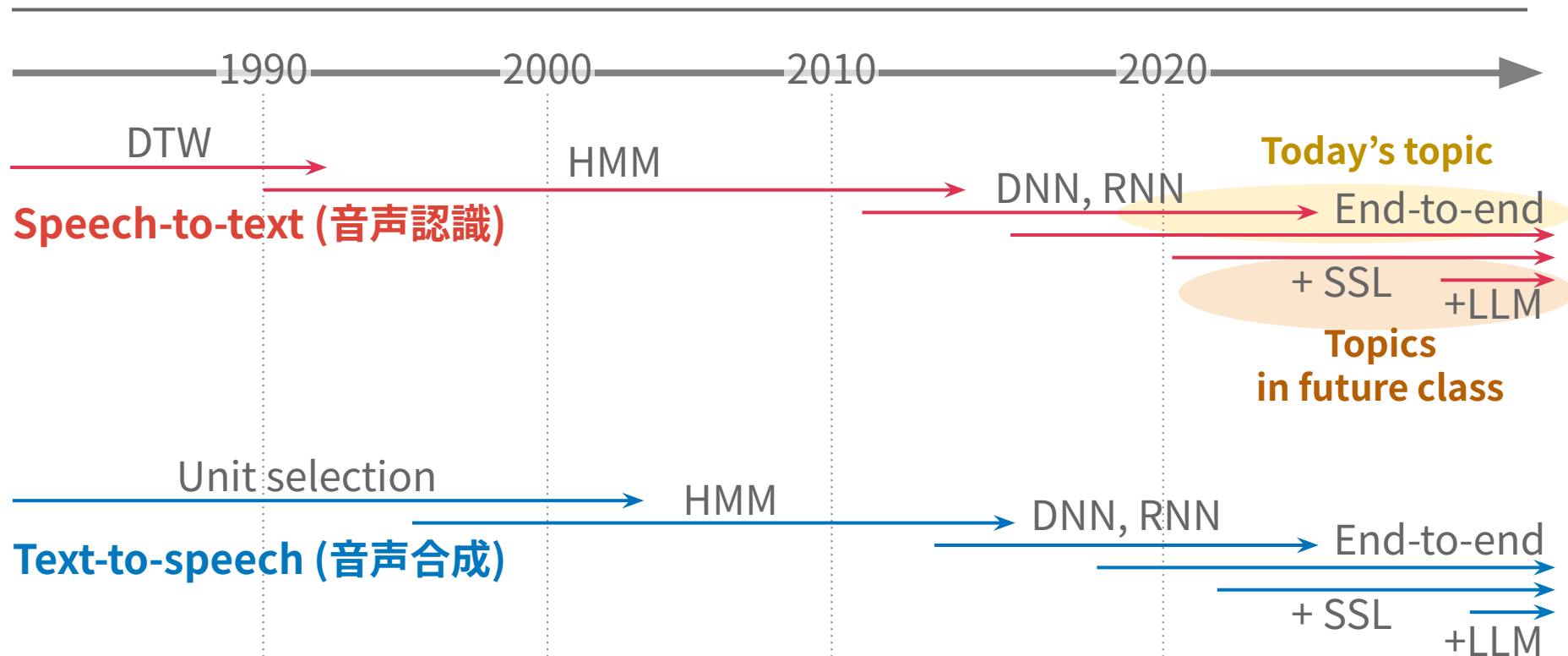
Prob. of linguistic feats.
given by word
(lexicon / 単語辞書)

Prob. of words.
(language model / 言語モデル)
Naturalness as text,
e.g., n-gram, GPT

Traditional flow (top) and end-to-end flow (bottom)



History



History

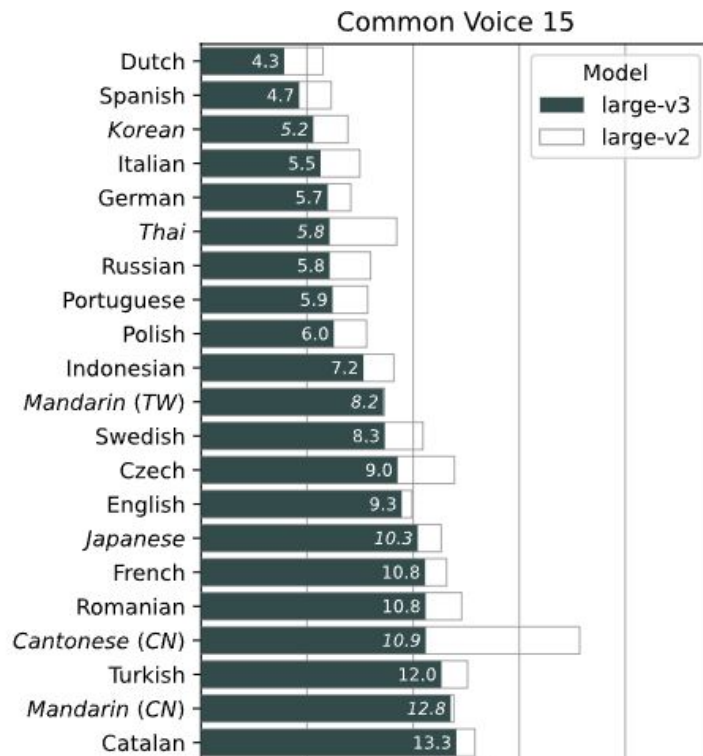
Phonetic typewriter (1962)

L1. Phonetic Typewriter. T. SAKAI (nonmember) AND S. DOSHITA (nonmember), *Kyoto University, Kyoto, Japan.*—We have constructed a research model of phonetic typewriter named Sonotype since 1960. This equipment was designed based upon our research results of these several years, aiming at both universal and flexible ability in the processing of speech sound. The basic equipment was a device for one hundred Japanese monosyllables in which about 3000 transistors and about 5000 diodes were used. For the purpose of treatment of Japanese conversational speech, some parts of control system have been added considering the data obtained from the basic equipment; however, the experimental result has not yet been obtained. This equipment has several peculiarities together with some features similar to those reported in other papers. (a) All the control pulses are derived from the input speech sound wave itself. (b) Distinctive features of speech sound are obtained by using not only the results of ordinary spectrum analysis but also various kinds of features necessary for the recognition of phonemes: for instance, voiced, unvoiced, fricative, nasal, contracted, and so on. (c) Discrimination is made by digital technique such as order pulse, AND gate, OR gate and binary circuit. Of course it is necessary to convert analog quantity to digital one and vice versa.

あ, い, ...ん, が,ぎ,...,きや,...

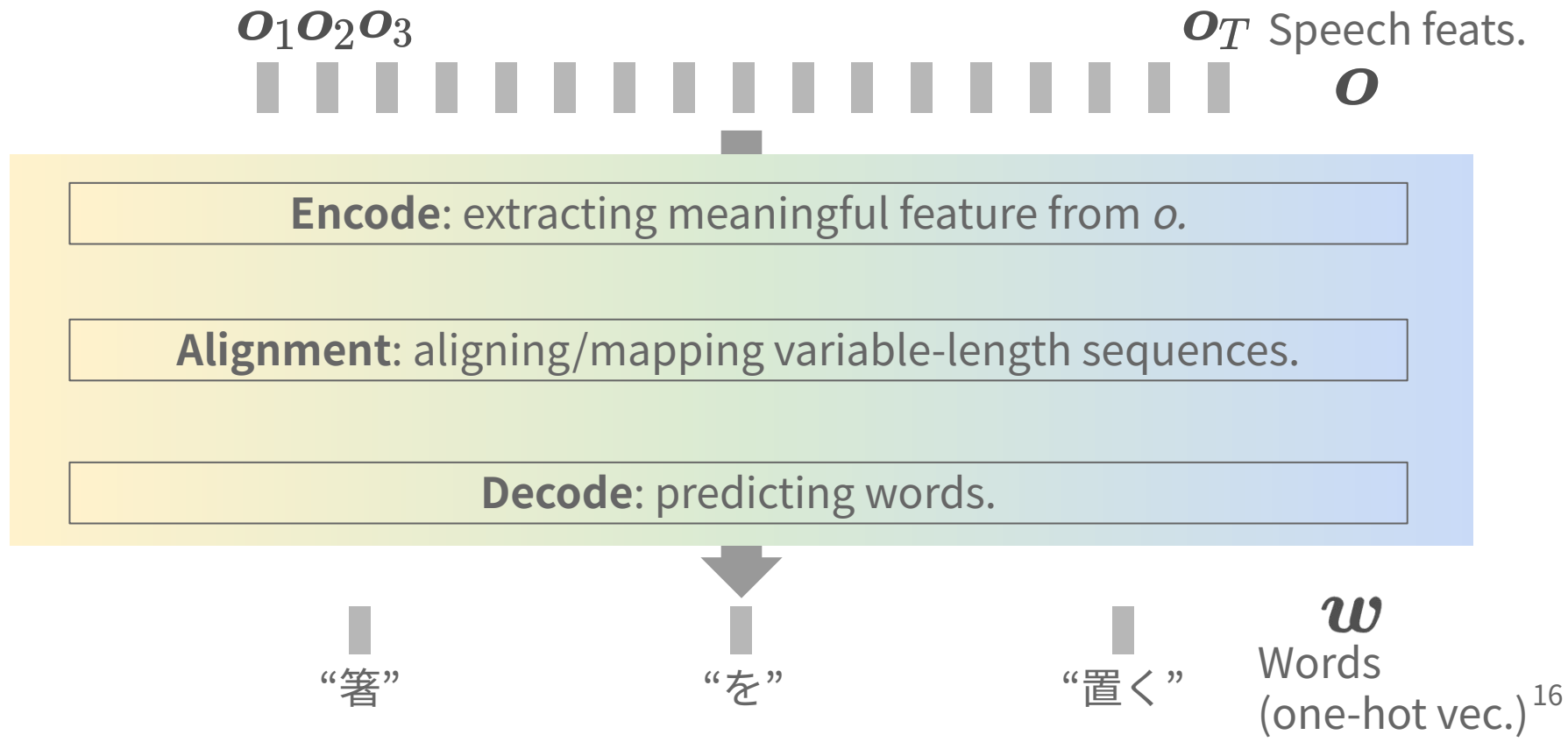
“one hundred Japanese monosyllables in which about 3000 transistors and about 5000 diodes”

Whisper v3 (2024): ~100 langs.



End-to-end model in speech-to-text / 音声認識における end-to-end モデル

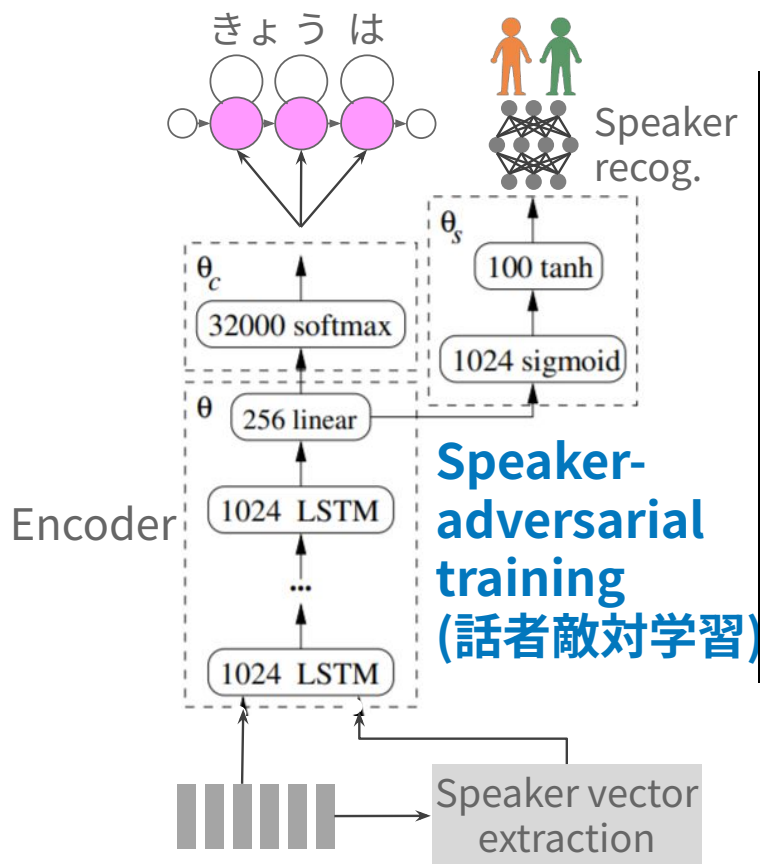
Into details: three inner modules



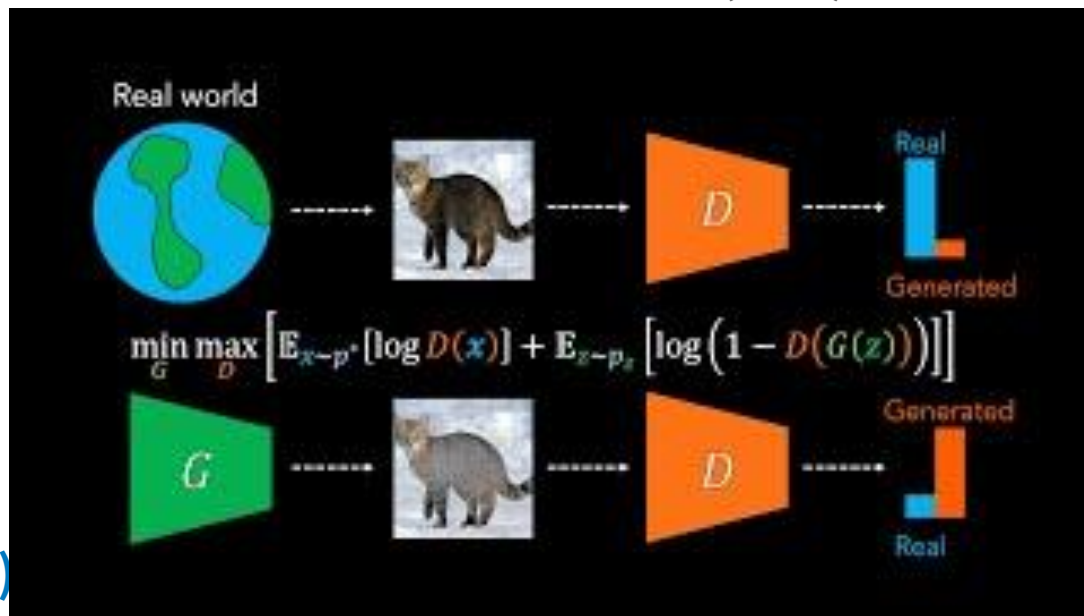
Encoder

- Extract meaningful features from speech.
- **Be sensitive to:**
 - Features to determine spoken text, phoneme, accent, prosody.
- **Be insensitive to:**
 - Speaker & their state: consistent among any speakers, diseases, etc.
 - Language: 7000+ languages, accented language (訛った言語)
 - straight (/s t r e i t/) vs. ストレート (/s **u** t **o** r e i t **o**/)
 - Environments: far/close, noise, reverberation, etc.

IBM speech-to-text (2017): achieving human parity in speech recognition



Generative adversarial network (GAN)



Speaker-adversarial training enforces the decoder on being insensitive to speakers.
-> able to extract speaker-independent features.

Alignment & Decoder

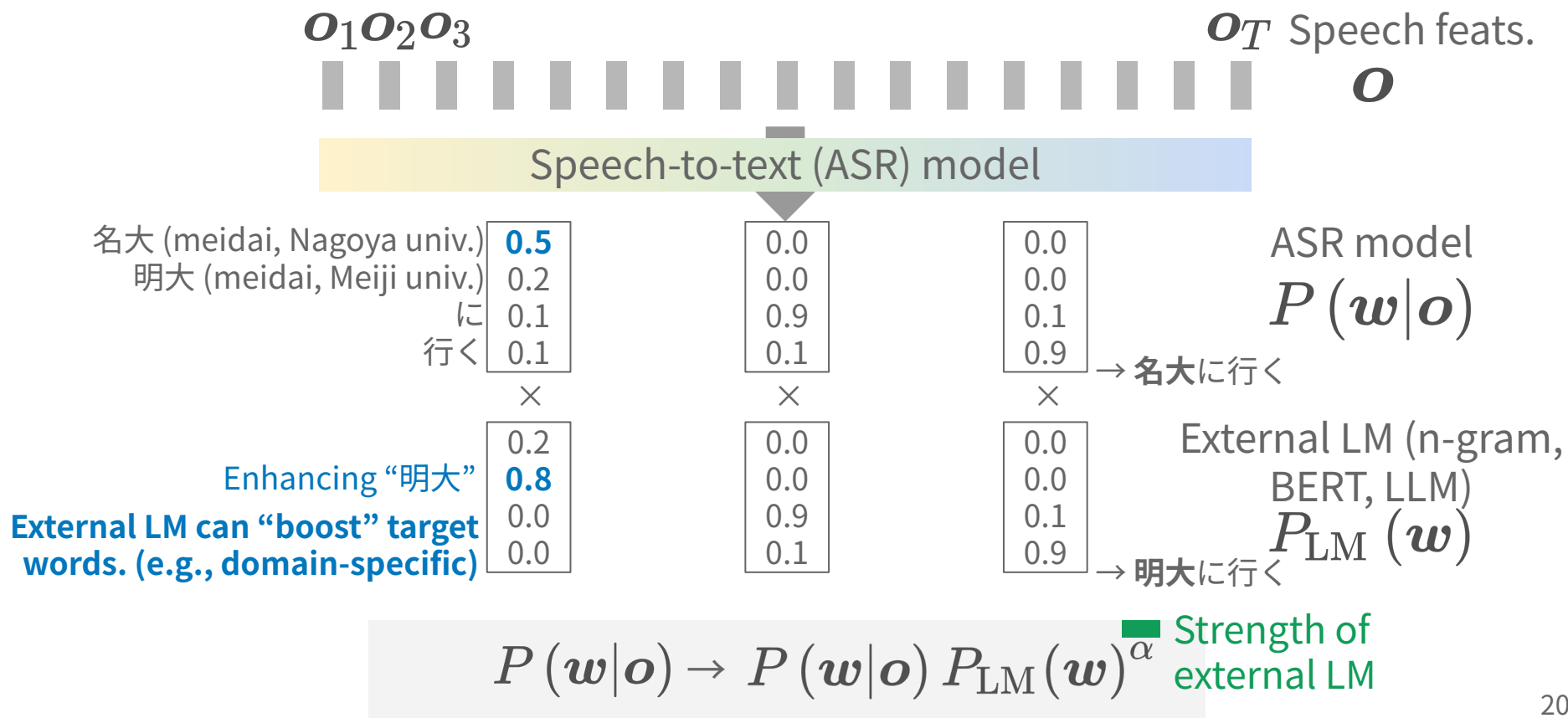
- Alignment: Soft vs. hard
 - Soft alignment: **hard computation** ($O(N^2)$) but **rich feature**
 - Hard alignment: **poor feature** but **light computation** (e.g., streaming)



- Decoder: Autoregressive vs. non-autoregressive
 - Autoregressive: **sequential computation** but **good prediction**
 - Current word can be predicted from past predicted words.
 - Non-autoregressive: **poor prediction** but **parallel computation** (i.e., fast)



(Advanced) external language model (LM) for decoder / (発展内容) 外部言語モデルの利用



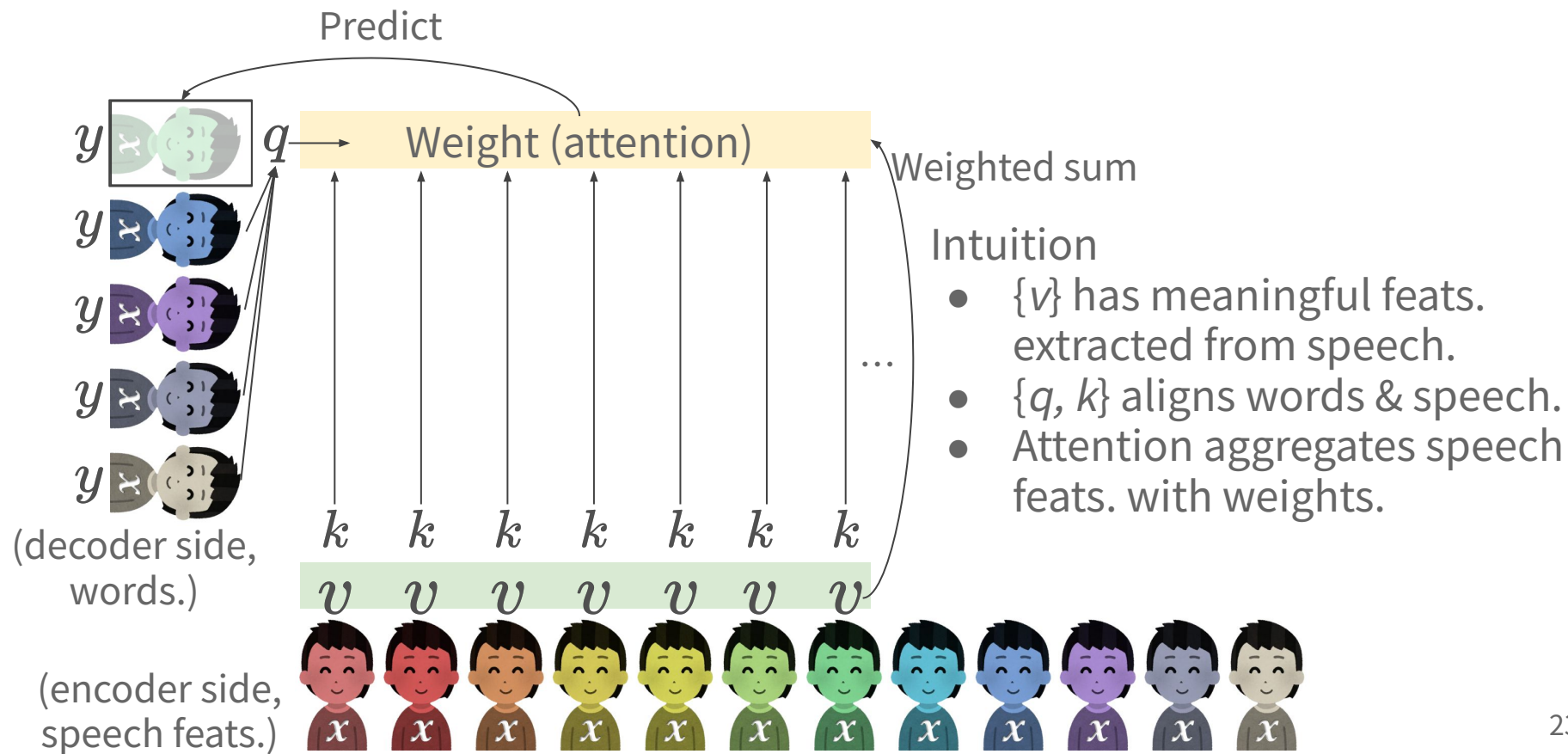
Three types of end-to-end model

	CTC (connectionist temporal classification)	Transducer	LAS (listen, attend, and spell)
Fig.	きょ_う_は_ きょ_う_は_ きょう_は_ Cross-entropy Use blank mark “_” to align sequences.	Current word “う” Prev. word “きょ” Recognize a word using prev.-recognized words.	きょ う は Query Predict Key & Value a.k.a., source-target attention
Pros.	HMM-like lightweight streaming process	Dependency among words & streaming process	Dependency among words & well-known technique
Cons.	Independence among words	Heavy computation in training (much memory)	Sequence-wise process

LAS (listen, attend and spell)

Encode Attention Decode

Cross-attention again



Whisper is based on LAS

Multitask training data (680k hours)

English transcription

- 🗣️ "Ask not what your country can do for ..."
- 📄 Ask not what your country can do for ...

Any-to-English speech translation

- 🗣️ "El rápido zorro marrón salta sobre ..."
- 📄 The quick brown fox jumps over ...

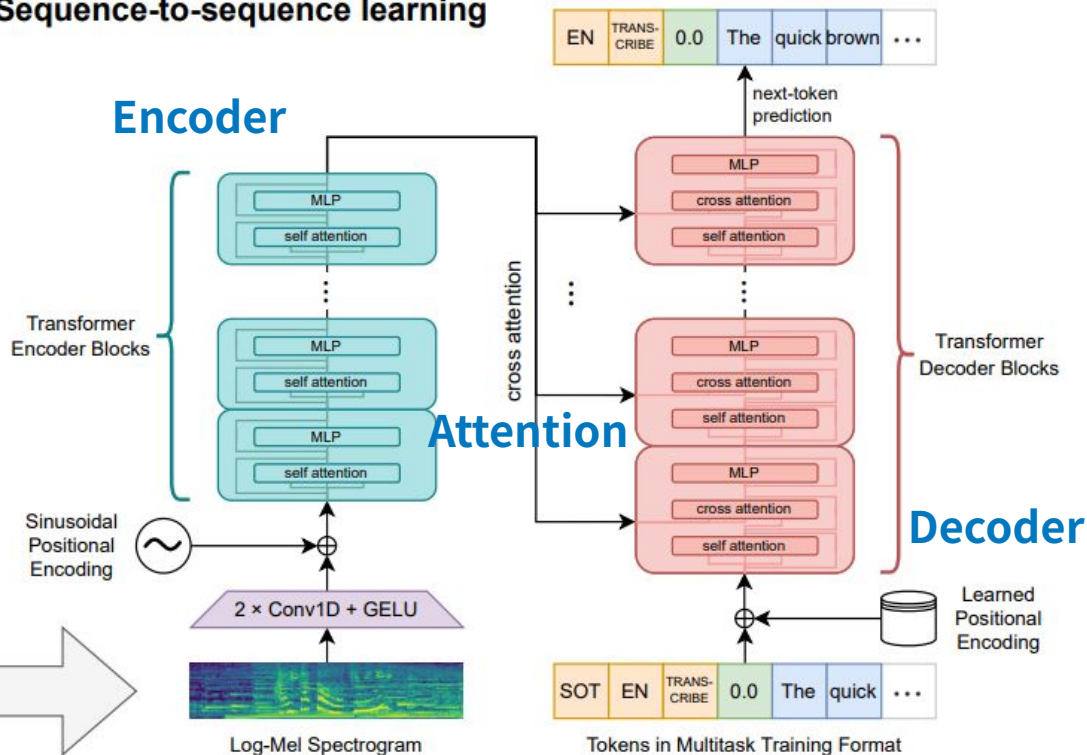
Non-English transcription

- 🗣️ "언덕 위에 올라 내려다보면 너무나 넓고 넓은 ..."
- 📄 언덕 위에 올라 내려다보면 너무나 넓고 넓은 ...

No speech

- 🔊 (background music playing)
- 📄 ∅

Sequence-to-sequence learning

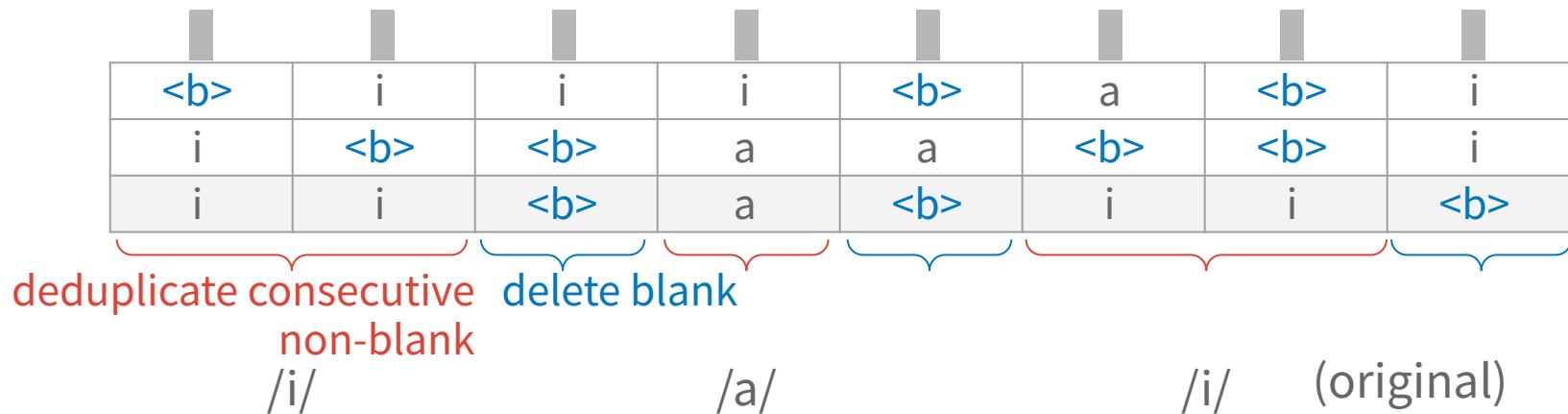


(Uniqueness of the Whisper is not to employ LAS but to adopt multi-task training.)

CTC (connectionist temporal classification)

Introducing “<blank>” (or “_”, “”) symbol / ブランク記号の導入

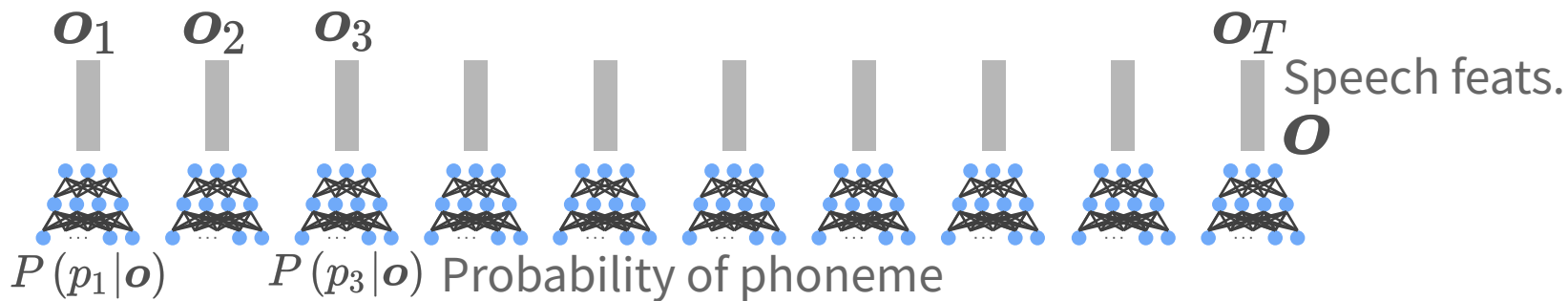
- blank = “no label”
 - Inserted so that the word (or phoneme) sequence has the same length to speech feature sequence
- Example (using phoneme but same when using words)
 - Assuming speech feature length is 8.
 - Extended sequence is designed to uniquely reconstruct the original.



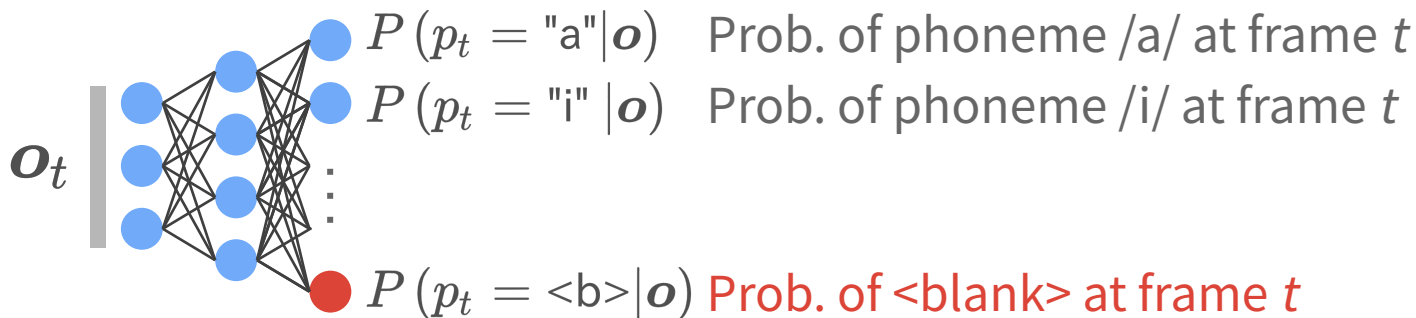
Output layer for CTC

/ CTC を用いるときの最終層

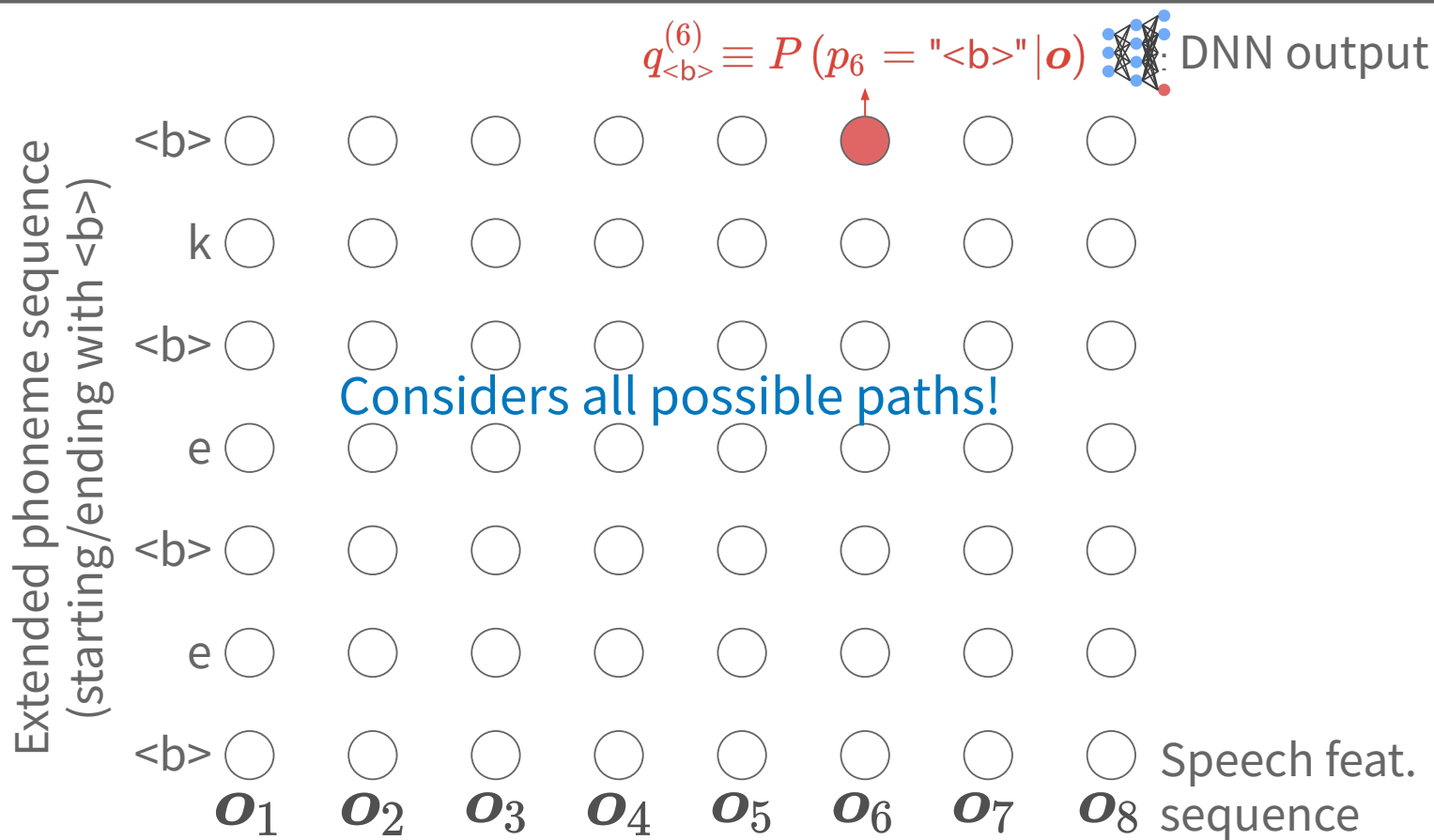
- Whole architecture



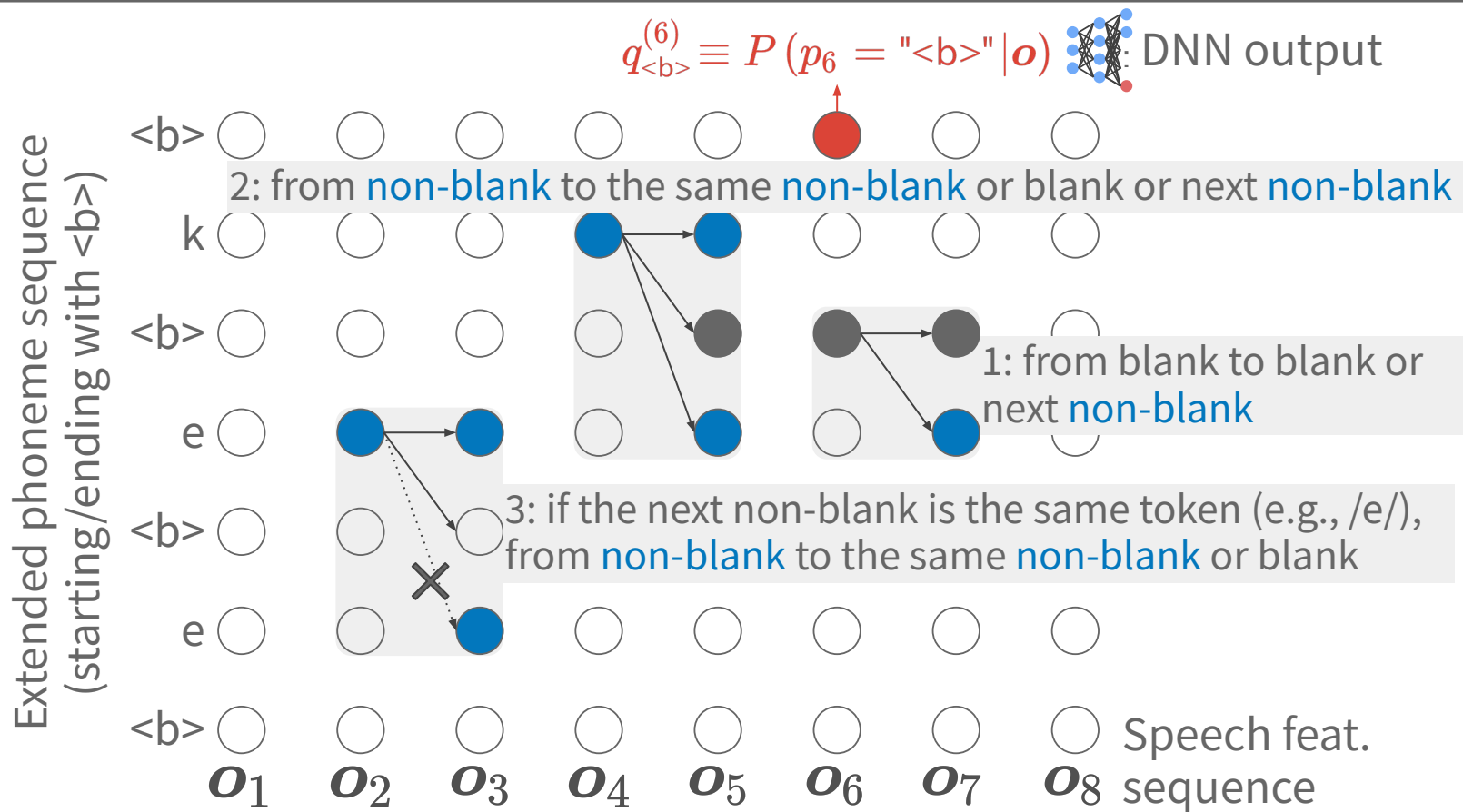
- Phoneme set + {blank}



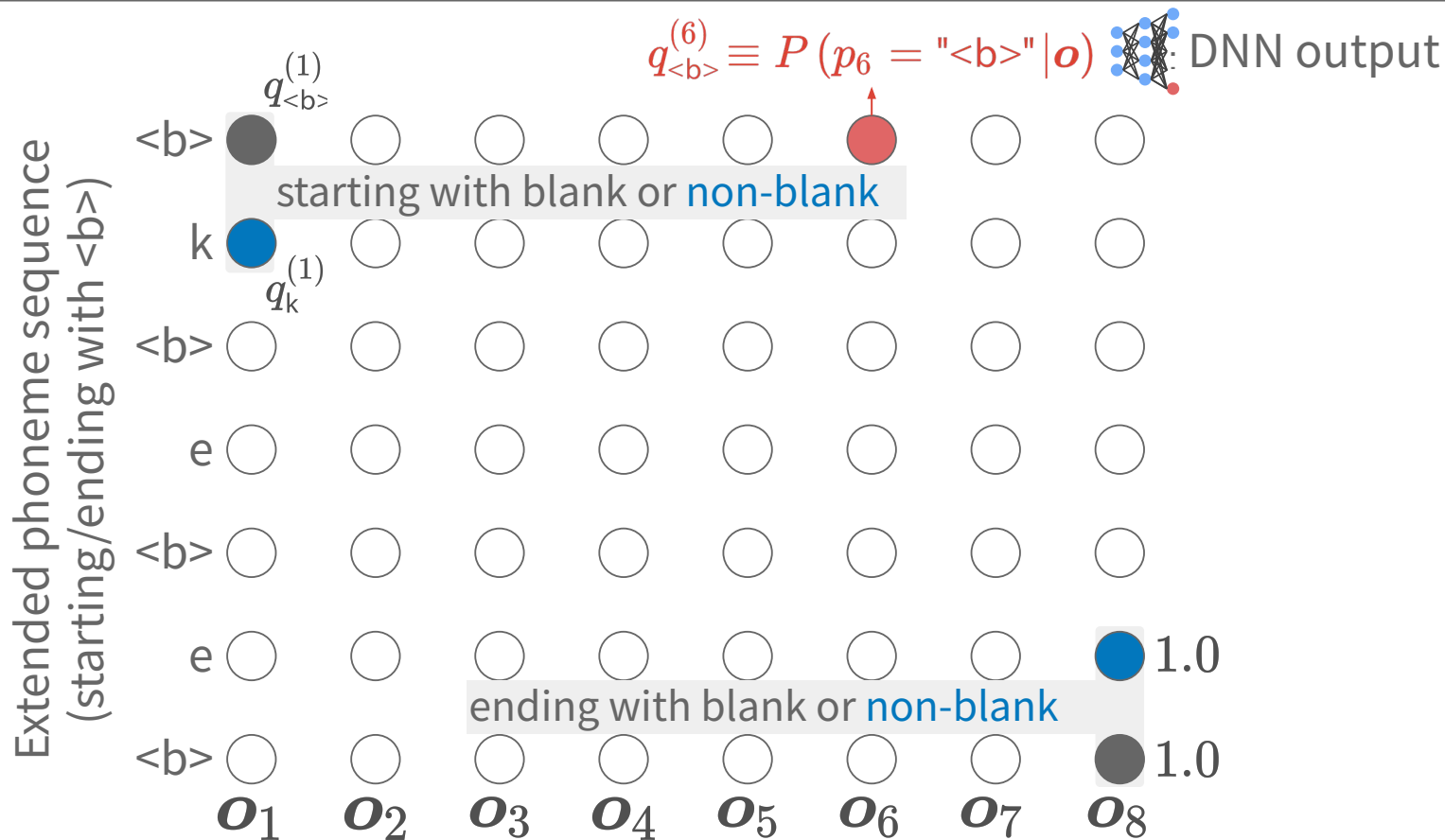
Alignment between speech feature sequence and extended phoneme sequence



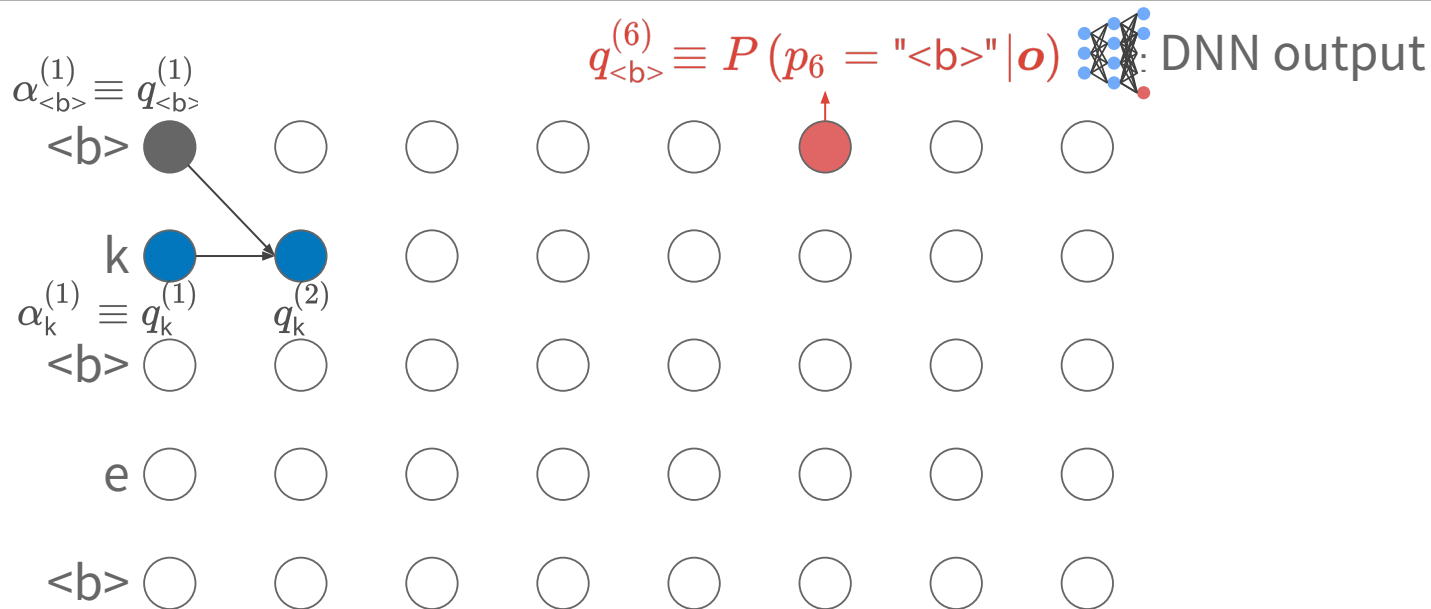
Transition conditions



Start-end conditions



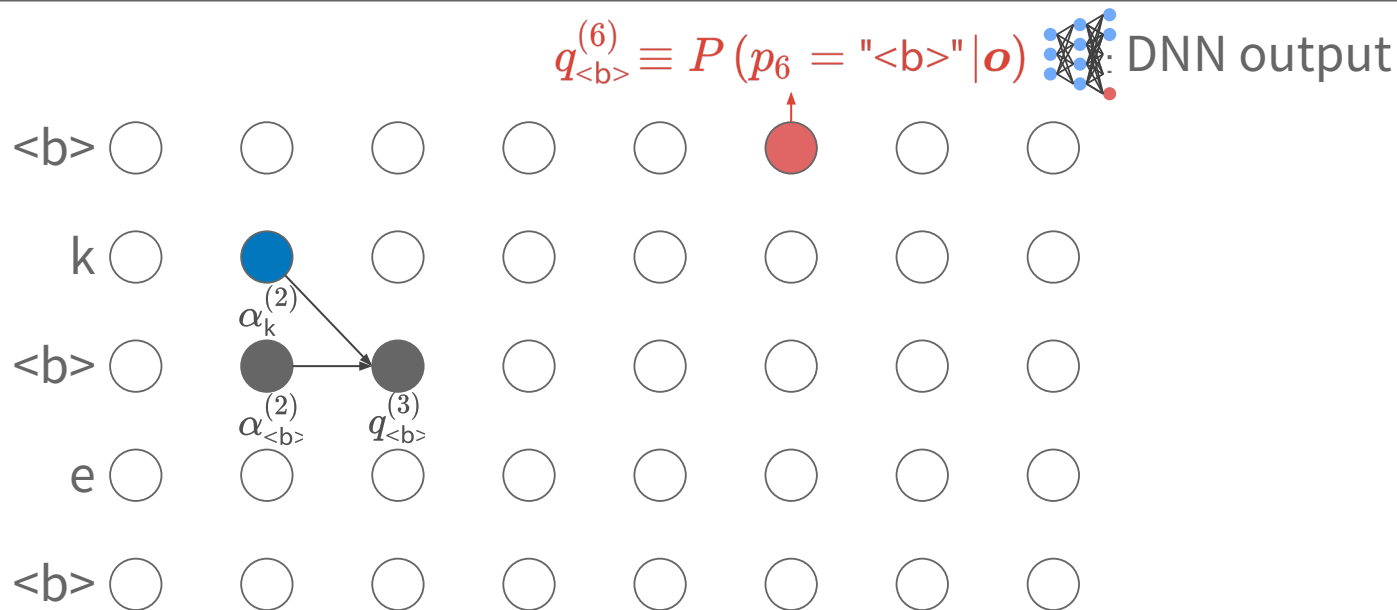
Forward transition



Cumulative score $\alpha_k^{(2)} = \left(\alpha_{}^{(1)} + \alpha_k^{(1)} \right) q_k^{(2)}$

$$ $\mathbf{o}_1$ $\mathbf{o}_2$ $\mathbf{o}_3$ $\mathbf{o}_4$ $\mathbf{o}_5$ $\mathbf{o}_6$ $\mathbf{o}_7$ $\mathbf{o}_8$

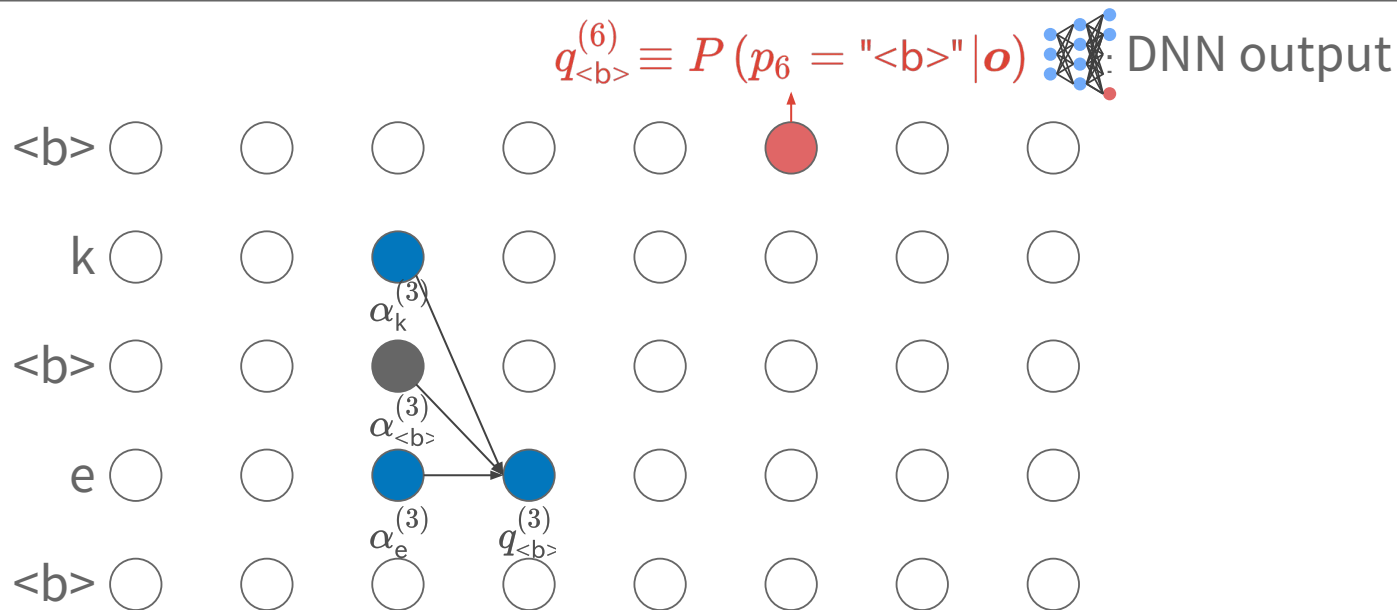
Forward transition



Cumulative score $\alpha_{}^{(3)} = \left(\alpha_k^{(2)} + \alpha_{}^{(2)} \right) q_{}^{(3)}$



Forward transition



Cumulative score $\alpha_e^{(4)} = \left(\alpha_k^{(3)} + \alpha_{}^{(3)} + \alpha_e^{(3)} \right) q_e^{(4)}$



Backward transition

$q_{}^{(6)} \equiv P(p_6 = "" | \mathbf{o})$  DNN output

 ○ ○ ○ ○ ○ ● ○ ○

Cumulative score $\beta_e^{(7)} = \beta_e^{(8)} q_e^{(8)} + \beta_{}^{(8)} q_{}^{(8)}$ ○ ○

 ○ ○ ○ ○ ○ ○ ○ ○

e ○ ○ ○ ○ ○ ○ ○ ○

 ○ ○ ○ ○ ○ ○ ○ ○

e ○ ○ ○ ○ ○ ● ● $\beta_e^{(8)} \equiv 1.0$

 ○ ○ ○ ○ ○ ○ ○ ● $\beta_{}^{(8)} \equiv 1.0$

$\mathbf{o}_1$ $\mathbf{o}_2$ $\mathbf{o}_3$ $\mathbf{o}_4$ $\mathbf{o}_5$ $\mathbf{o}_6$ $\mathbf{o}_7$ $\mathbf{o}_8$

Backward transition

$q_{}^{(6)} \equiv P(p_6 = "" | \mathbf{o})$  DNN output

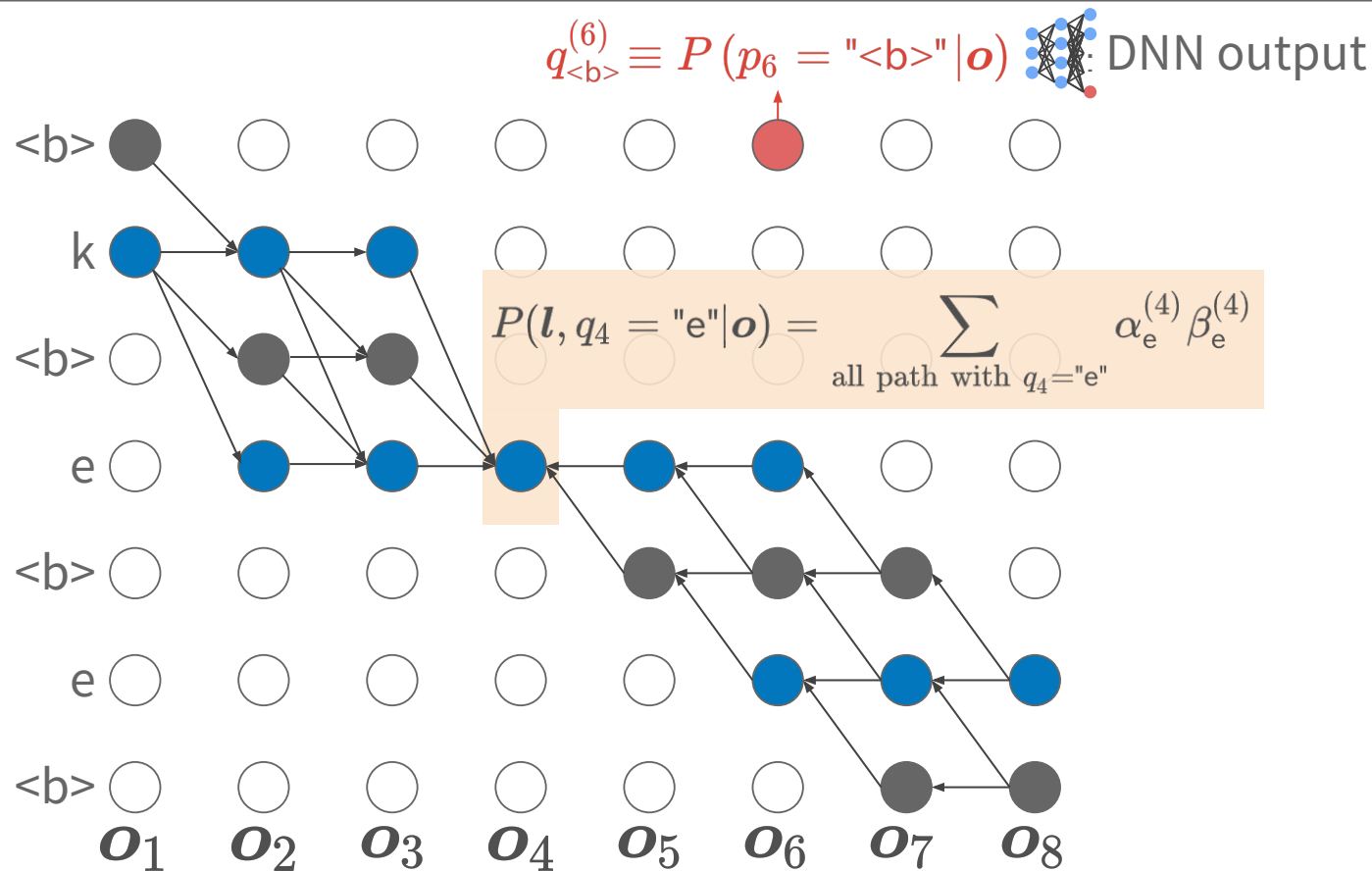


Cumulative score $\beta_e^{(4)} = \beta_e^{(5)} q_e^{(5)} + \beta_{}^{(5)} q_{}^{(5)}$



$\mathbf{o}_1$ $\mathbf{o}_2$ $\mathbf{o}_3$ $\mathbf{o}_4$ $\mathbf{o}_5$ $\mathbf{o}_6$ $\mathbf{o}_7$ $\mathbf{o}_8$

Possibility of a phoneme at a frame is calculated by marginalizing all possible paths with the phoneme.

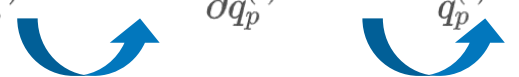


CTC training

- Objective function to be minimized

$$\mathcal{L}_{\text{CTC}} = -\log P(\mathbf{l}|\mathbf{o})$$

- Gradient (for backpropagation) at phoneme p , time t .

$$\frac{\partial \mathcal{L}_{\text{CTC}}}{\partial q_p^{(t)}} = -\frac{1}{P(\mathbf{l}|\mathbf{o})} \frac{\partial P(\mathbf{l}|\mathbf{o})}{\partial q_p^{(t)}}$$
$$\frac{\partial P(\mathbf{l}|\mathbf{o})}{\partial q_p^{(t)}} = \frac{\partial P(\mathbf{l}, q_t = p|\mathbf{o})}{\partial q_p^{(t)}} = \frac{1}{q_p^{(t)}} P(\mathbf{l}, q_t = p|\mathbf{o})$$


Remove
constant terms.

$P()$ is a linear
function of $q_p^{(t)}$.

$$\frac{\partial \mathcal{L}_{\text{CTC}}}{\partial q_p^{(t)}} = -\frac{1}{P(\mathbf{l}|\mathbf{o})} \frac{1}{q_p^{(t)}} P(\mathbf{l}, q_t = p|\mathbf{o})$$

You: aggregating all possible paths are very complicated. The backpropagation is definitely complicated!

Me: No! Once you estimated the probability, the gradient can be simply estimated!

CTC inference

- Greedy search / 貪欲法
 - Find the most possible phoneme or blank at each frame.
 - Convert the extended phoneme sequence to no-blank sequence.
 - **Simplest** but assuming **local best (maximum)**.
 - Set of local best is not equivalent to the global best (maximum).
- Beam search / ビームサーチ
 - One kind of breadth-first search / 幅優先探索
 - **Keep N -best possible paths** and determine the best after searching all frames.
 - See <https://www.youtube.com/watch?v=V-VEilUoWuI> for the detail.
 - “CTC Decoding and Beam Search”
 - **Can find better path than greedy search** (but requires more computation)₈

Evaluation / 評価

How we evaluate ASR hypothesis?

/ どうやって音声認識結果を評価する？

- Example (I am studying in the University of Tokyo.)
 - Reference (正解)：私は東京大学で学んでいます
 - Hypothesis (認識結果)：私は統計学を学んでいます
(I am studying statistics.)
- Metrics
 - SER (sentence error rate) / 文誤り率
 - WER (word error rate) / 単語誤り率
 - CER (character error rate) / 文字誤り率

Metrics

- SER (sentence error rate)

- The sentence is totally correct or not (0% or 100%)

Ref.) 東京大学で学んでいます

Hyp.) 統計学を学んでいます

SER = 100 %

- WER (word error rate)

- Word-level edit distance (編集距離)

Ref.) 東京/大学/で/学ん/で/い/ます

Hyp.) 統計学/を/学ん/で/い/ます

Insertion error (挿入誤り) : 0

Deletion error (削除誤り) : 1

Substitution error (置換誤り) : 2

Ref. length : 7

WER = $(2 + 1) / 7 * 100$ [%]

- CER (character error rate)

- Character-level edit distance

Ref.) 東/京/大/学/で/学/ん/で/い/ま/す

Hyp.) 統/計/学/を/学/ん/で/い/ま/す

Calculated in the same manner as
in WER

Off-topic: orthography (正書法)

- A set of norms for spelling, capitalization, punctuation, etc.
/ スペルや句読点など、言語を記述するときの規範の集合
 - Different among languages: “center” vs. “centre”
 - {SER, WER, CER} supposes the orthography in the target language.
/ {SER, WER, CER} は正書法の存在を仮定している。
- Many exceptions / たくさんの例外
 - Indic languages, e.g., Hindi
 - devanāgarī / デーヴァナーガリー, Urdu / ウルドゥ文字
 - Japanese
 - Hiragana, Katakana, Kanji (Chinese characters)
 - Any of {たばこ, タバコ, 煙草} is acceptable.
(/t a b a k o/, smoking)

WER calculation based on Levenshtein distance

/レーベンシュタイン距離に基づく WER の計算

		仮説 (Hypothesis) $\rightarrow i$			
$j \downarrow$ 見本 (Reference)			大学	へ	行く
	きょう				
	カーニバル		?		
	へ				
	行く				

WER is the edit distance of two sequences of diff. lengths that have been aligned.

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

仮説 (Hypothesis)

→ i

		大学	へ	行く
--	--	----	---	----

$d(i, j)$ = $\min (d[i - 1, j] + 1, \blacksquare \text{ insertion error / 挿入誤り}$

distance of
(i, j)

$d[i, j - 1] + 1, \blacksquare \text{ deletion error / 削除誤り}$

$d[i - 1, j - 1] + \text{Local}(i, j)) \blacksquare \text{ substitution error / 置換誤り}$

1 when $\text{word}_i = \text{word}_j$, otherwise 0

行く				
----	--	--	--	--

↓
 j

WER is the edit distance of two sequences of diff. lengths that have been aligned.

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

		仮説 (Hypothesis)				→ i
見本 (Reference) ↓ j			大学	へ	行く	
		0	→ 1	→ 2	→ 3	
	きょう					
	カーニバル					
	へ					
	行く					

$$\min(\underbrace{d[i-1, j] + 1}_{\text{Insertion / 挿入}}, \underbrace{d[i, j-1] + 1}_{\text{Deletion / 削除}}, \underbrace{d[i-1, j-1] + \text{Local}(i, j)}_{\text{Substitution / 置換}})$$

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

		仮説 (Hypothesis)				→ i
見本 (Reference) ↓ j			大学	へ	行く	
		0	1	2	3	
	きょう	↓1				
	カーニバル	↓2				
	へ	↓3				
	行く	↓4				

$$\min (\underbrace{d[i-1, j] + 1}_{\text{Insertion / 挿入}}, \underbrace{d[i, j-1] + 1}_{\text{Deletion / 削除}}, \underbrace{d[i-1, j-1] + \text{Local}(i, j)}_{\text{Substitution / 置換}})$$

Insertion / 挿入

Deletion / 削除

Substitution / 置換

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

		仮説 (Hypothesis) → i			
見本 (Reference) ↓ j			大学	へ	行く
		0	1	2	3
	きょう	1	$\min(2, 2, 1)$		
	カーニバル	2			
	へ	3			
	行く	4			

$\min(\underbrace{d[i-1, j] + 1}_{\text{Insertion / 挿入}}, \underbrace{d[i, j-1] + 1}_{\text{Deletion / 削除}}, \underbrace{d[i-1, j-1] + \text{Local}(i, j)}_{\text{Substitution / 置換}})$

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

		仮説 (Hypothesis) → i			
見本 (Reference) ↓ j			大学	へ	行く
		0	1	2	3
	きょう	1	1		
	カーニバル	2	$\min(3, 2, 2)$		
	へ	3			
	行く	4			

$\min(\underbrace{d[i-1, j] + 1}_{\text{Insertion / 挿入}}, \underbrace{d[i, j-1] + 1}_{\text{Deletion / 削除}}, \underbrace{d[i-1, j-1] + \text{Local}(i, j)}_{\text{Substitution / 置換}})$

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

		仮説 (Hypothesis)				→ i
見本 (Reference) ↓ j			大学	へ	行く	
		0	1	2	3	
	きょう	1	1	2		
	カーニバル	2	2	2		
	へ	3	3	$\min(4, 3, 2)$	Local(i, j)=0	
	行く	4	4			

$\min(\underbrace{d[i-1, j] + 1}_{\text{Insertion / 挿入}}, \underbrace{d[i, j-1] + 1}_{\text{Deletion / 削除}}, \underbrace{d[i-1, j-1] + \text{Local}(i, j)}_{\text{Substitution / 置換}})$

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

		仮説 (Hypothesis) → i			
見本 (Reference) ↓ j			大学	へ	行く
		0	1	2	3
	きょう	1	1	2	3
	カーニバル	2	2	2	3
	へ	3	3	2	3
	行く	4	4	3	2

$$\min (\underbrace{d[i-1, j] + 1}_{\text{Insertion / 挿入}}, \underbrace{d[i, j-1] + 1}_{\text{Deletion / 削除}}, \underbrace{d[i-1, j-1] + \text{Local}(i, j)}_{\text{Substitution / 置換}})$$

Insertion / 挿入

Deletion / 削除

Substitution / 置換

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

		仮説 (Hypothesis)				→ i
見本 (Reference) ↓ j			大学	へ	行く	
		0	1	2	3	
	きょう	1	1	2	3	
	カーニバル	2	2	2	3	
	へ	3	3	2	3	
	行く	4	4	3	2	

The shortest path is their alignment.

(Reference length = 4, insertion = 0, deletion = 1, substitution = 1)

WER calculation based on Levenshtein distance

/レーベンシュタイン距離に基づく WER の計算

- WER (CER) may exceed 100%. / WER (CER) は 100%を超える場合あり
 - E.g., Many insertion and deletion.
- Alignment path is not unique. / アライメントは一意には決まらない
 - Multiple shortest paths may exist. In that case, one path is selected on some criteria. / 最短経路が複数存在する場合がある。複数存在する場合、なんらかの基準で1つ選ぶことが多い
- Check not only WER but also inner errors. / 各誤りを確認
 - Many insertion errors ... failed to detect end of speech.
 - Many deletion errors ... much background noises.

Conclusion / まとめ

Conclusion

- Formulation of speech-to-text (ASR)
- End-to-end model for ASR
 - CTC, transducer, and attention
- Evaluation
 - SER, WER, CER